# Kahlan Hagerman

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## ABOUT ME

Final-year Computer Science student at the University of Pretoria. Backend-focused experience with TypeScript, NestJS, Supabase, and PostgreSQL, plus systems-level programming in C++ and Java. Comfortable working across the stack, from backend services and databases to personal systems projects, with a strong grounding in software design and problem-solving.

## EDUCATION

**University of Pretoria** 2024 – Present
*BSc Computer Science* Pretoria, South Africa

- **Relevant coursework**: Computer Graphics, Artificial Intelligence, Programming Languages, C++ Design Patterns, Databases, Concurrent Programming, Security

## PROJECTS

**SpendSense** Capstone Project (COS301), 2026

- Led backend API development with **NestJS and Supabase**, including auth/RBAC and Jest/Supertest test coverage
- Primarily backend, with some frontend work; gamified credit score simulator with streaks, badges, leaderboards, and quizzes; set up the project board and README

**Design Patterns Project** Group Project (COS214)

- Implemented 13+ GoF design patterns (Composite, Observer, State, Factory Method, Abstract Factory, Command, Iterator, and more) in a C++ plant nursery management system with a Raylib GUI
- Wrote Google Test unit tests, Doxygen documentation, and UML diagrams; repository

**Raycasting Engine** Personal Project (C++)

- Built a raycasting engine from scratch in C++ using Raylib for real-time rendering
- Implemented a multi-floor elevator system, inventory mechanics, and ambient audio

**standup-cli** Personal Project

- Built a TypeScript/Node.js CLI that summarises git commit history using local AI (Ollama, Llama 3.2) via Commander.js

**Municipal Services Platform** Mini Project (COS301)

- Implemented accounts API endpoints on **TypeScript/Express, Supabase, Next.js**, tested with Jest/Supertest and Puppeteer

## SKILLS

**Languages** : TypeScript, JavaScript, C++, Java, SQL, HTML/CSS
**Frameworks & Tools** : Node.js, Express, NestJS, Supabase, PostgreSQL, React, Git/GitHub, Jest, Docker